Enhancing Text-to-Image Alignment with Retrieval-Augmented GPT for Historical Event Reconstruction: Evaluating with Multimodal LLMs

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Introduction

A picture is worth a thousand words, particularly in history, where visuals complement text to deepen understanding of past events (Dussel & Priem 2017). E.g., visualizing a Roman triumph can animate ancient rituals in ways that textual descriptions alone cannot. Beyond enhancing engagement (Dehouche & Dehouche 2023), such visualizations offer valuable tools for scholarly research, enabling historians to model and interpret the spatial, performative, and political dimensions of historical phenomena more concretely. While some scholars criticize visual media in historical narratives (Depaepe & Henkens 2000), others argue that images function as semiotic systems comparable to language (Peim 2005), underscoring their analytical potential in reconstructing and contextualizing the past.

(a) **Model:** DALL-E 3, **Prompt:** *"Imagine ancient Rome during Iulius Caesars time. Caesar is sitting on a gilded chair atop the Rostra in the roman forum. He is wearing a triumphal attire and a crown on his head. The forum is crowded with Roman citizens and we can see the roman architecture."*

(b) **Model:** SDXL, **Prompt:** *"Show an image of Marcus Antonius as lupercus during the Lupercalia in 44 BCE, just before he wants to offer the diadem, which he hides behind his back, to Caesar. Caesar sits in his sella aurea, wearing the toga purpurea and the corona aurea."*

(c) **Model:** FLUX.1-dev, **Prompt:** *"Imagine the triumphal procession of the Roman Emperor Probus in a carriage pulled by six horses in 281 CE. Probus wears the usual triumphal regalia. He holds a wand and laurel in his hands. The carriage is accompanied by two soldiers."*

Figure 1: Failed cases from DALL-E 3, SDXL, and FLUX.1-dev, with wrongly represented elements.

Recent advances in text-to-image (T2I) models (Yu et al. 2022, Rombach et al. 2022, Saharia et al. 2022, Ramesh et al. 2022, Betker et al. 2023) have improved generative quality, but aligning complex prompts with accurate visuals remains challenging. Misrepresentations, such as incorrect attire or architecture, are common (Figure 1), especially when prompts involve rare cultural or temporal references.

Manual prompt engineering is time-consuming and requires domain-specific expertise (Gal et al. 2023). For example, accurately describing a Roman general's attire or the Lupercalia festival can easily overwhelm non-experts. This paper addresses these challenges by introducing an automatic prompt refinement approach using retrieval-augmented GPT-4o (OpenAI 2024), which improves historical text-to-image generation through evidence-based alignment. By lowering the barrier to producing accurate visualizations, this method enables a wider range of scholars to incorporate visual representations into their research and teaching, thus expanding the use of AI-generated imagery as a tool for historical inquiry.

Our Contributions

1. A historical domain prompt-image dataset comprising 200 prompts and 6,000 images by five T2I models.
2. An automatic T2I alignment evaluation and assessment of evaluation consistency with three MLLMs.
3. A human evaluation of a subset.

Related Work

Prompt Engineering

Prompt engineering designs instructions to guide pre-trained models in generating outputs without modifying core parameters. T2I models typically employ (1) prompt tuning and (2) prompt optimization. Prompt tuning (Jia et al. 2022) adapts models with task-specific templates and few-shot examples, but scaling to large models becomes computationally intensive. Prompt optimization methods (e.g., *Promptist* (Hao et al. 2023)) refine user prompts to enhance image aesthetics. Building on these, our approach focuses on improving T2I alignment rather than aesthetics by leveraging an LLM for prompt refinement. We integrate domain-specific knowledge into prompts using retrieval-augmented generation (RAG) (Lewis et al. 2020), enabling historically accurate image generation.

Parameter-Efficient Fine-Tuning (PEFT) for Image Models

PEFT methods (Ding et al. 2023) adapt large-scale models for downstream tasks by updating only a subset of parameters. Techniques such as Low-Rank Adaptation (LoRA) (Hu et al. 2022) and adapter-based tuning (Houlsby et al. 2019) enable efficient customization. Originally developed for LLMs, these methods are now widely applied to T2I diffusion models (Saharia et al. 2022, Rombach et al. 2022, Hu et al. 2022, Mou et al. 2024, Sohn et al. 2023). Our work employs LoRA to fine-tune T2I models on a small dataset, balancing efficiency and performance to improve alignment in historical contexts.

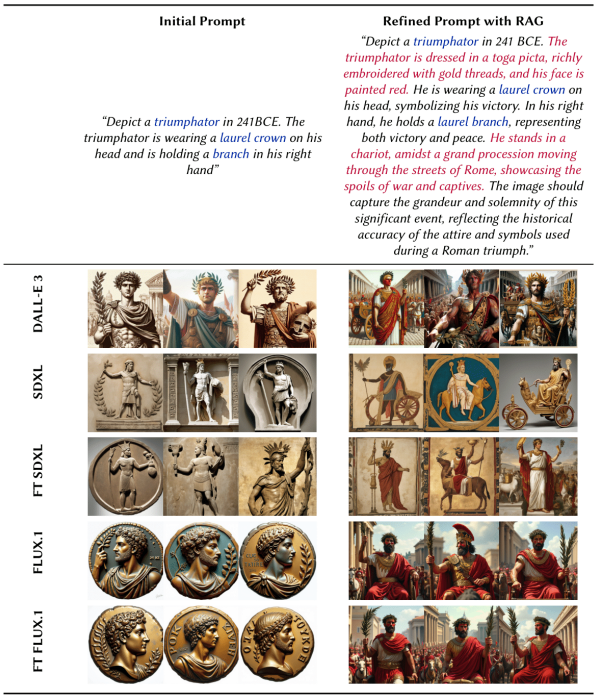

Figure 2: Comparison of triumph images (3 random samples) generated by five T2I models using initial and refined prompts; main objects used for retrieval, added historical context.

Dataset Preparation

Our dataset was prepared in two steps:

1. **Prompt Generation** We used 100 manually crafted prompts from (Ströbel et al. 2024), comprising 75 depicting triumphal processions in ancient Rome and 25 centered on the *Lupercalia* festival.
2. **Image Model Fine-tuning and Image Generation** We generated 3,000 images using five models: DALL-E 3 (Betker et al. 2023), SDXL (Podell et al. 2024), fine-tuned SDXL, FLUX.1-dev,[1] and fine-tuned FLUX.1-dev.[2] Each prompt from step 1 was used to generate six images. For fine-tuning, we curated 276 caption-image pairs, of which 177 pairs are from two historical books (Gorski & Packer 2015b, Kiley 2013) on architecture, clothing, and weapons, and 99 are frames from the TV series Rome, S1E10,[3] automatically captioned with Claude 3.5 Sonnet.[4] We fine-tuned on Replicate using LoRA (Hu et al. 2022) and textual inversion (Gal et al. 2023) for SDXL, and Ostris's AI Toolkit for FLUX.1-dev. We applied three lora_scale values (0.2, 0.4, 0.6), generating two images per value for each prompt with the fine-tuned mo-

dels. The complete dataset will be publicly available on GitHub.[5]

RAG with GPT-4o

The goal of this study is to enhance T2I alignment through automatic prompt refinement using RAG via OpenAI's Assistant API,[6] leveraging the Chain-of-Thought prompting strategy (Wei et al. 2022).

GPT-4o extracts historical objects from initial prompts within predefined categories and uses them as queries to retrieve contextual content from a historical domain database.[7] The retrieved content enriches the prompts, enabling better alignment with the historical objects described.

The experiment refined the 100 human-generated prompts, generating six images per refined prompt across five T2I models, yielding 3,000 images. We applied the same lora_scale settings from the initial dataset generation for fine-tuned models. Appendix A Figure 6 provides detailed instructions on the refinement process.

Evaluation

T2I alignment evaluates how well an image's visual semantics match a prompt's textual semantics (Yarom et al. 2023). We adopted two evaluation criteria from (Ströbel et al. 2024): (1) QG-VQA using MLLMs across the full dataset and (2) a 5-point Likert scale evaluation with both MLLMs and human raters on a subset. The MLLMs used were GPT-4o, Gemini 1.5 Pro (Reid et al. 2024), and Claude 3.5 Sonnet.

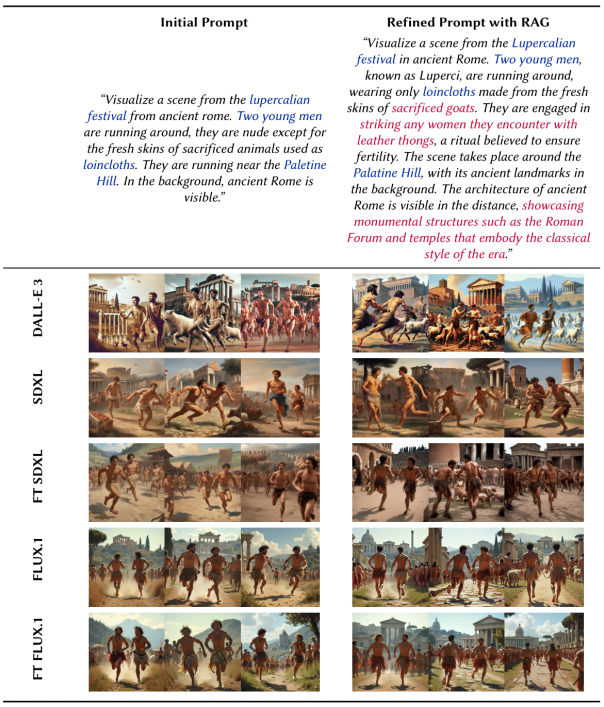

Figure 3: Comparison of Lupercalia images (three random samples) from five T2I models using initial and refined prompts; main objects used for retrieval, added historical context.

QG-VQA Setup

The QG-VQA metric (Hu et al. 2023, Cho et al. 2024) involved generating questions with GPT-3.5 (Brown et al. 2020) and answering them using the MLLMs. Binary responses (1 for "Yes," 0 for "No") were averaged per image to compute a final alignment score ranging from 0 (no alignment) to 1 (perfect alignment). See Appendix A Figure 8 for the prompt.

Prompts	Models	GPT VQA_{SD}		Claude VQA_{SD}		Gemini VQA_{SD}	
		Triumph	*Lupercalia*	Triumph	*Lupercalia*	Triumph	*Lupercalia*
Init	DALL-E 3	$\mathbf{0.65}_{0.25}$	$\mathbf{0.65}_{0.23}$	$0.10_{0.11}$	$0.14_{0.14}$	$0.49_{0.25}$	$0.44_{0.21}$
Refined	DALL-E 3	$0.35_{0.18}$	$0.44_{0.18}$	$0.16_{0.12}$	$\mathbf{0.18}_{0.14}$	$\mathbf{0.60}_{0.18}$	$\mathbf{0.54}_{0.20}$
Init	SDXL	$0.30_{0.24}$	$0.24_{0.16}$	$0.17_{0.14}$	$0.15_{0.12}$	$0.39_{0.24}$	$0.35_{0.22}$
Refined	SDXL	$0.28_{0.16}$	$0.37_{0.17}$	$0.10_{0.09}$	$0.11_{0.09}$	$0.42_{0.19}$	$0.37_{0.20}$
Init	finetuned SDXL	$0.40_{0.27}$	$0.32_{0.19}$	$\mathbf{0.20}_{0.16}$	$\mathbf{0.18}_{0.14}$	$0.41_{0.24}$	$0.33_{0.20}$
Refined	finetuned SDXL	$0.32_{0.15}$	$0.45_{0.18}$	$0.14_{0.11}$	$0.17_{0.12}$	$0.48_{0.20}$	$0.39_{0.20}$
Init	FLUX.1	$0.36_{0.26}$	$0.24_{0.19}$	$0.06_{0.08}$	$0.09_{0.09}$	$0.32_{0.21}$	$0.28_{0.18}$
Refined	FLUX.1	$0.35_{0.17}$	$0.38_{0.19}$	$0.13_{0.10}$	$0.15_{0.11}$	$0.42_{0.18}$	$0.33_{0.19}$
Init	finetuned FLUX.1	$0.16_{0.15}$	$0.18_{0.16}$	$0.16_{0.18}$	$0.09_{0.09}$	$0.32_{0.21}$	$0.27_{0.16}$
Refined	finetuned FLUX.1	$0.38_{0.16}$	$0.26_{0.15}$	$\mathbf{0.20}_{0.11}$	$0.15_{0.12}$	$0.43_{0.18}$	$0.31_{0.19}$

Table 1: VQA-based T2I alignment evaluation.

Prompts	Models	Rater-A Likert		Rater-B Likert		Rater-C Likert	
		Triumph	*Lupercalia*	Triumph	*Lupercalia*	Triumph	*Lupercalia*
Init	DALL-E 3	**3.4**	3.2	**2.8**	**2.6**	2.9	2.4
Refined	DALL-E 3	3.2	**3.6**	1.9	2.2	**3.0**	**3.4**
Init	SDXL	1.7	2.2	1.5	1.6	1.9	1.6
Refined	SDXL	1.8	1.8	1.1	1.0	2.0	1.8
Init	finetuned SDXL	1.8	2.2	1.9	2.0	2.0	1.8
Refined	finetuned SDXL	2.0	2.4	1.5	1.8	2.0	1.8
Init	FLUX.1	2.7	2.2	3.0	1.8	2.4	2.2
Refined	FLUX.1	2.4	1.8	2.2	2.4	2.6	2.6
Init	finetuned FLUX.1	2.2	2.4	2.5	2.0	2.1	2.2
Refined	finetuned FLUX.1	2.2	1.4	2.4	2.4	2.7	2.2

Table 2: Human 5-point Likert scale evaluation.

Prompts	Models	Rater-GPT Likert		Rater-Claude Likert		Rater-Gemini Likert	
		Triumph	*Lupercalia*	Triumph	*Lupercalia*	Triumph	*Lupercalia*
Init	DALL-E 3	**4.3**	3.6	**4.2**	3.8	**4.2**	3.6
Refined	DALL-E 3	4.0	**4.6**	3.7	**4.2**	4.1	**4.0**
Init	SDXL	2.9	2.8	3.5	3.6	2.8	3.0
Refined	SDXL	3.4	3.2	4.0	3.8	2.1	1.8
Init	finetuned SDXL	3.1	3.2	3.8	4.0	3.8	3.8
Refined	finetuned SDXL	3.2	3.0	**4.2**	**4.2**	3.1	3.2
Init	FLUX.1	2.7	3.2	3.3	3.2	3.4	2.8
Refined	FLUX.1	2.7	3.0	3.5	3.8	3.7	3.0
Init	finetuned FLUX.1	2.5	3.2	3.6	3.6	3.3	3.2
Refined	finetuned FLUX.1	3.3	2.8	3.4	3.4	3.4	2.8

Table 3: MLLM 5-point Likert scale evaluation.

5-point Likert Scale Setup

We conducted a Likert scale evaluation on 150 randomly sampled prompt-image pairs, with 15 pairs per T2I model (10 from Triumph and 5 from *Lupercalia*). Human and MLLM raters assessed alignment on a scale from 1 (no match) to 5 (perfect match). Refined prompt-image pairs were evaluated alongside their initial counterparts, and Appendix A Figure 7 details the evaluation guidelines.

Results

Table 1 presents the average VQA scores for five T2I models: DALL-E 3, SDXL, fine-tuned SDXL, FLUX.1-dev, and fine-tuned FLUX.1-dev, using initial and refined prompts. Refined prompts consistently achieved higher VQA scores, particularly for DALL-E 3, FLUX.1, and fine-tuned FLUX.1, with the largest improvement observed for FLUX.1, which achieved a 0.22-point increase. Tables 2 and 3 detail the 5-point Likert scale evaluations by humans and MLLMs, respectively. Refined prompts notably improved alignment in the *Lupercalia* category for DALL-E 3, although performance varied in the Triumph category.

LoRA Fine-Tuning Enhances SDXL Performance. The results indicate that LoRA fine-tuning improves SDXL's T2I alignment across both VQA and Likert-scale evaluations. Fine-tuned SDXL consistently outperformed its base version, achieving the largest improvement in the Triumph category. As Tables 2 and 3 show, human *and* MLLM assessments favored fine-tuned SDXL, confirming the effectiveness of combining textual inversion with LoRA for modeling intricate historical concepts.

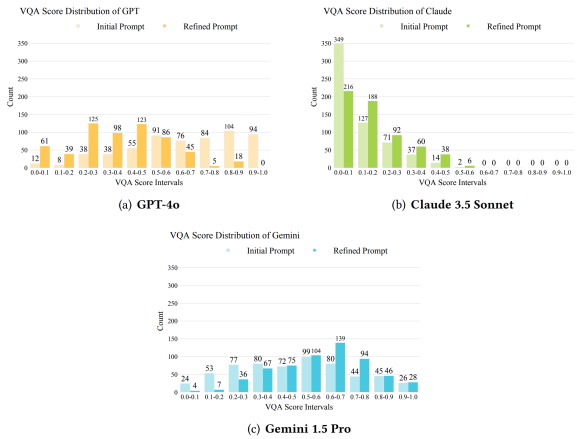

Figure 4: VQA score distribution from three MLLMs for DALL-E 3 images using initial (lighter saturation) and refined (darker saturation) prompts.

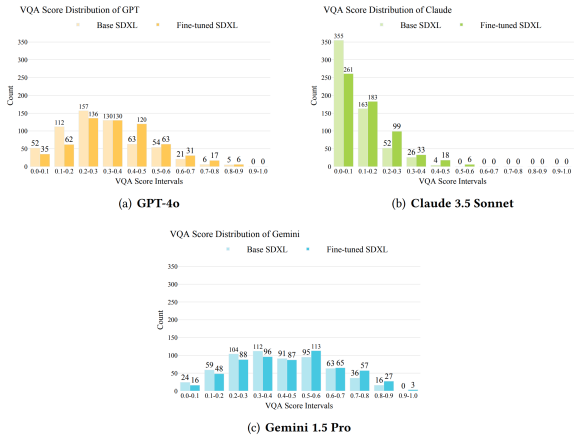

Figure 5: VQA score distribution from three MLLMs for images generated with refined prompts using base (lighter saturation) and fine-tuned (darker saturation) SDXL.

Rating Divergences Among MLLMs. Figures 4 and 5 show notable differences in VQA scores across MLLMs. GPT-4o and Gemini exhibited similar scoring patterns, typically assigning mid-range scores (0.3–0.7), while Claude's ratings predominantly clustered in the lower range (0.0–0.3). Jensen-Shannon divergence analysis indicated high similarity between GPT-4o and Gemini's rating distributions for DALL-E 3 and SDXL, while Claude's scores diverged significantly. These findings are supported by inter-annotator agree-

ment (IAA) results in Table 6, which highlight stronger alignment between GPT-4o and Gemini but lower overall agreement across all MLLMs.

Refined Prompts Lead to Better Object Alignment. As Figures 2 and 3 illustrate, refined prompts consistently produced images that better aligned with the described objects, especially in historically complex scenes. E.g., refined prompts improved depictions of Roman architectural details in Lupercalia, while initial prompts often failed to capture these nuances.

Evaluation Complexity. The IAA results in Table 4 reflect significant discrepancies in VQA evaluations among MLLMs, with all scores falling below the 0.67 threshold. These results highlight the inherent complexity of evaluating T2I alignment and underscore the need for more robust and interpretable metrics.

Prompts	Models	IAA		
		GPT vs. Gemini	GPT vs. Claude	Gemini vs. Claude
Init	DALL-E 3	0.31	-0.78	-0.66
Refined	DALL-E 3	-0.16	0.09	-0.54
Init	SDXL	0.32	-0.45	-0.56
Refined	SDXL	-0.04	-0.04	-0.42
Init	finetuned SDXL	0.35	-0.51	-0.58
Refined	finetuned SDXL	0.02	-0.06	-0.34
Init	FLUX.1	0.31	-0.45	-0.53
Refined	FLUX.1	0.11	-0.07	-0.29
Init	finetuned FLUX.1	0.11	-0.37	-0.53
Refined	finetuned FLUX.1	0.37	0.11	0.03

Table 4: Krippendorff's # between MLLM raters' VQA scores.

Conclusion

This study demonstrates the potential of retrieval-augmented GPT-4o to enhance T2I alignment by refining prompts with historical context. Refined prompts improved alignment across five T2I models (Tables 3 and 4), with LoRA fine-tuning further enhancing SDXL performance. These results emphasize the importance of integrating historical specificity into prompt engineering.

Evaluation discrepancies among MLLMs (Table 6) highlight the need for more robust metrics, while GPT-4o's occasional hallucinations during refinement affect reliability. The small human evaluation dataset limits generalizability. Future work should explore re-ranking strategies and larger datasets to address these challenges.

By bridging gaps in T2I alignment, this work advances the generation of historically accurate visualizations, paving the way for deeper insights into historical events.

Acknowledgements

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Appendix A – Additional Figures

```
"""
You are tasked with refining a historical image generation prompt to improve its
alignment with historical accuracy and reduce noisy, irrelevant elements.

Follow these steps:
1.  First, you will be presented with an initial prompt for historical image
generation.
2.  Extract the main historical objects or elements from the initial prompt.
These should be key items (such as architecture, clothing, uniforms, weapons,
and military equipment), people, or concepts that are central to the historical
scene being described. List these objects in <historical_objects> tags.
3.  For each historical object you've identified, retrieve relevant historical
information.  This may include details about its appearance, context, or
significance during the specific time period mentioned in the prompt. Summarize
this information briefly for each object within <historical_context> tags.
4. Using the initial prompt and the historical context you've gathered, refine
the prompt to make it more historically accurate and focused. Your refined prompt
should:
  - Emphasize the key historical elements
  - Include accurate details from your research
  - Remove any anachronistic or historically inaccurate elements
  - Provide clear visual descriptions that align with the historical period
5. Present your refined prompt within <refined_prompt> tags.
6. Your entire output should be in XML format, structured as follows:
  <output>
    <historical_objects>[List of main historical objects/
elements]</historical_objects>
    <historical_context>[Summary of historical information for each
object]</historical_context>
    <refined_prompt>[Your refined, historically accurate
prompt]</refined_prompt>
  </output>

Remember to focus on historical accuracy and relevance, ensuring that the refined
prompt will generate an image that is true to the intended historical period and
context.
"""
```

Figure 6: Prompt for contextual information retrieval and image prompt refinement with GPT-4oAssistant API using CoT.

```
"""
You are tasked with evaluating the alignment between a historical image from
ancient Rome and a given prompt.  Your goal is to assess how well the image
matches the prompt using a 5-point Likert scale.  Follow these instructions
carefully to provide a thorough and accurate evaluation.

First, carefully analyze the provided historical image.

Now, consider this prompt that the image was supposed to match: {image_prompt}

Analyze the visual elements in the image in relation to the prompt. Consider how
well the image captures the elements specified in the prompt. Pay close attention
to the following aspects:
1. Setting and background:
  - Does the environment match the description in the prompt?   - Are there any
anachronistic or out-of-place elements?
2. Characters or figures:
  - Are all the characters mentioned in the prompt present in the image?
  - Do they match the descriptions given (e.g., appearance, clothing, status)?
3. Poses, actions, and interactions:
  - Do the characters' poses and actions align with what's described in the
prompt?
  - Are the interactions between characters accurate to the prompt?
4. Objects and artifacts:
  - Are all objects mentioned in the prompt present in the image?
  - Do they appear historically accurate for ancient Rome?
5. Historical accuracy:
  - Are there any elements that seem out of place for the time period of ancient
Rome?
6. Unspecified elements:
  - Are there any significant elements in the image that were not mentioned in
the prompt?
  - If so, do these additions enhance or detract from the alignment with the
prompt?
7. Missing elements:
  - Are there any important elements from the prompt that are not represented in
the image?

Based on your analysis, evaluate the image on the following 5-point Likert scale:
  1. The image does not match the prompt at all.
  2. The image barely contains aspects of the prompt.
  3. The image captures some aspects of the prompt, but it is not very accurate.
  4. The image captures most of the aspects of the prompt.
  5. The image completely matches the prompt.

Present your evaluation in the following format:
<score>[Provide the numerical score (1-5) here]</score>

Remember to be thorough in your analysis and provide a fair, well-justified
evaluation based on all aspects of the image-prompt alignment.
"""
```

Figure 7: Prompt for 5-point Likert scale evaluation.

```
"""
You are an expert in historical imagery, specializing in ancient Roman history.
Your task is to analyze a given historical image and answer a question about it.
It is crucial that you provide only a Yes or No answer.

The question you need to answer about this image is: {question}

Please follow these steps:
  1. Carefully examine the image, paying close attention to details that might
be relevant to ancient Roman history.
  2. Consider the question in relation to what you observe in the image.
  3. Based on your expertise in ancient Roman history and the visual information
provided, determine whether the answer to the question is affirmative or negative.

Remember, you must answer ONLY with Yes or No
Do not provide any explanation or additional information beyond this single-word
response.
"""
```

Figure 8: Prompt for VQA with GPT-4o, Claude 3.5 Sonnet, and Gemini 1.5 Pro.

Notes

1.
2. Fine-tuned via Replicate: https://replicate.com .
3.
4.
5.
6.
7. 31 documents in three categories: (1) Roman architecture (Hales 2003, Miles 2016, Southern 2006, Kleiner 2014, Anderson 2013, Marconi 2015, Müller et al. 2005, Tolqin 2021, Sear 2002, Lott 2018, Ulrich & Quenemoen 2013, Gorski & Packer 2015a), (2) clothing (Matz 2001, Garland 2009, Olson 2012, Fögen & Lee 2009, Noy 2002, Peachin 2011, Boatwright 2012, De Ligt & Northwood 2008, Lo Cascio 2018, Montserrat 2002, Parani 2008, Bonfante & Sebesta 2001, Noy 2018), and (3) the Roman army (Erdkamp 2007, Tacoma 2016, Coulston 2018, Barker 2015, Hassall 2000, Kiley 2012).

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